WASP-35b

R-1634

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-35_171106` · created 2026-08-02T06:08:23+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458063.897 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.013
Transit depth (Rp/Rs)^2	2.39 +/- 0.41 [%]
Orbital Inclination (inc)	86.48 +/- 1.42
Airmass coefficient 1 (a1)	0.988 +/- 0.019
Airmass coefficient 2 (a2)	0.008 +/- 0.025
Scatter in the residuals of the lightcurve fit is	1.92 %
Best Comparison Star	#1 - [358, 205]
Optimal Method	PSF photometry
Transit Duration (day)	0.122 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.013 vs 0.1241 (deviation 1.6σ)
Duration fit vs published	0.122 d vs 0.1317 d (deviation 0.47σ)
Mid-time O–C	1.1 min
Depth SNR	8.0
Residual scatter	1.92 %

Light curve

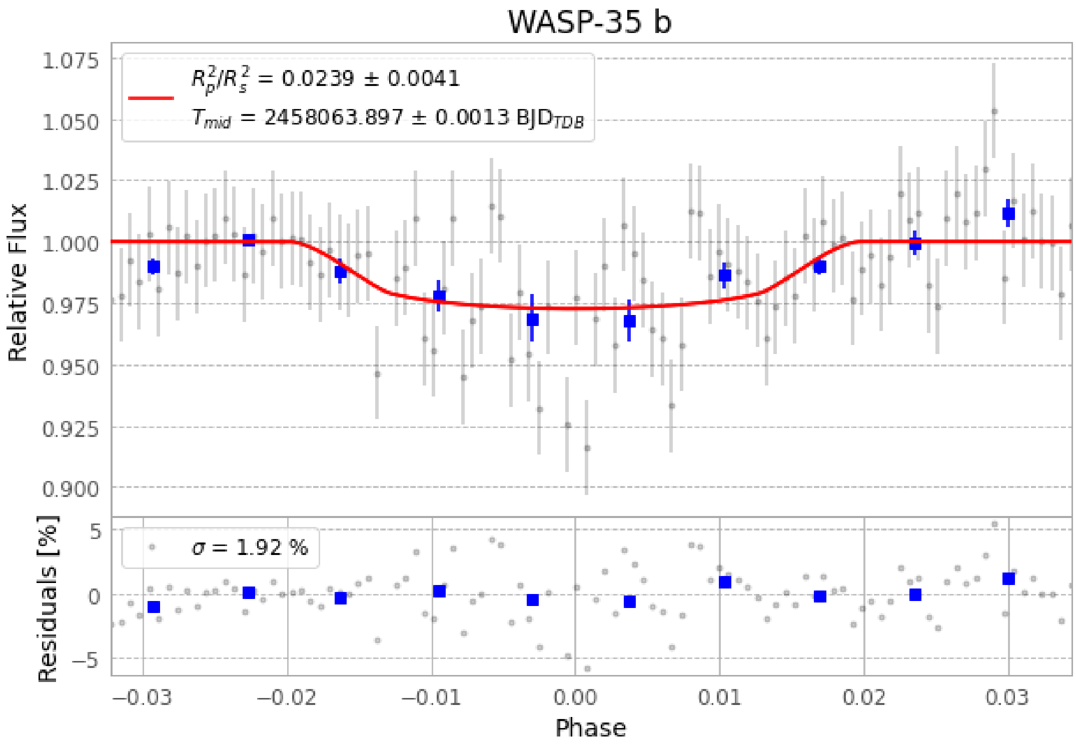

Figure 1 — FinalLightCurve_WASP-35 b_2017-11-05.png (SHA-256 243c604e621ecd70…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [274, 238], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 2b3477851792b05c…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

102 FITS frames (67 MB), `WASP-35171106065712.FITS` → `WASP-35171106120012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-171106.log (bff33b6ae763…), FinalLightCurve_WASP-35 b_2017-11-05.png (243c604e621e…), FinalLightCurve_WASP-35 b_2017-11-05.csv (50671a38ec37…)

Compute resources

Wall time 245.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-08-02 06:08Z.